

6th national RISC-V student contest 2025-2026

Sponsored by Thales, the GDR SOC² and the CNFM

Guidelines to report results

This document is a complement to the **Annonce RISC-V contest 2025-2026 v1.3.pdf** file (and its subsequent versions) to prepare the submission of results.

1. Prerequisites

1.1 Verify your setting

The reference project is the one you can find in https://github.com/ThalesGroup/cva6-softcore-contest/tree/cv32a6_contest_25_26.

Before starting developing your solution, you should check that the reference project runs correctly in your environment:

- The kit is running in your Linux environment.
- You can launch all targets of the Makefile as described in the **README.md** file.
- The simulation of the FFT application runs to its end and returns a SUCCESS (meaning it returns the same result as the reference, on a per-bit basis).
- The FFT application executes on the Zybo Z7-20¹ board and returns a SUCCESS.
- The simulation of the CoreMark binary runs to its end and returns a performance result.
- The CoreMark application executes on the Zybo board.
- The simulation of the MNIST binary runs to its end and returns a successful result: the '4' digit is recognized and the credence is 82.
- The MNIST application executes on the Zybo board and reports the same result.

1.2 Verify the references

In addition, before working on your optimizations, it is wise to ensure that:

- You are using the same OS, tools and preferably the same tool versions² as in the **README.md** file.
- You get the same performance metrics with the reference project as the organization team.

¹ No other FPGA boards will be accepted as the jury will replay results on its board.

² This is a recommendation so that the organizers can provide support. You can use other tools, e.g. Verilator, but you won't receive support on these tools.

The reference metrics are:

Console output for the FFT in simulation:

```
# [UART]: FFT running...  
# [UART]: FFT finished  
# [UART]: kiss_fft took 124142 instructions and 153496 cycles  
# [UART]: SUCCESS : fft result values correct
```

For the FFT on the Zybo board :

```
FFT running...  
FFT finished  
kiss_fft took 124142 instructions and 169248 cycles  
SUCCESS : fft result values correct
```

The number of cycles on the board can slightly vary from one execution to the other.

For the MNIST in simulation:

```
Expected = 4  
Predicted = 4  
Result : 1/1  
credence: 82  
image env0003: 1760387 instructions  
image env0003: 2834951 cycles
```

We do not care about the MNIST performance in this year's contest, but you should get similar.

The key figures of the FPGA design are:

- Minimum clock period of the design as reported after place & route in the FPGA (not out-of-context): 24.906ns
- FPGA resources of your top-level design as reported in
corev_apu/fpga/report_cva6_fpga_impl/cva6_fpga.utilization.rpt:
 - o 21996 LUTs
 - o 16438 FFs
 - o 60 RAM36

If the results with the reference project are not the same as above, you first need to check that you are using the same versions of the tools. If you still have a difference of more than 2% after this, get in touch with the organizers.

2 Analyzing the results

Before submitting results for the contest, you need to ensure that you fulfill all constraints described in the **Announce RISC-V contest 2025-2026 v1.3.pdf** file (or its subsequent versions).

Even if your solution works in simulation, it is important to verify that it also **works on the Zybo board**.

2.1 FFT performance

In this year's competition, the goal is to reduce the number of cycles needed by a CV32A6-based processor to compute an FFT-512 by modifying CV32A6 architecture or adding a coprocessor.

The simulation testbench and the FPGA top design are a bit different, mostly with respect to main memory access time. Therefore you can experience different numbers of cycles between both environments. The jury will consider the **number of cycles in simulation** to rank the teams. But your solution shall also work on the board.

2.2 Frequency

If there are ties, i.e. teams with a close number of cycles (<1%), the results will be separated with the maximum frequency as measured with Vivado timing analysis.

Also, you cannot decrease the CV32A6 operational frequency more than 20%.

You can determine the maximal clock frequency of your design with the static timing analysis report after the place & route:

Timing Report

```
Slack (MET) : 0.094ns (required time - arrival time)
Source: i_ariane/i_cva6/csr_regfile_i/pmpaddr_q_reg[6][1]/C
(rising edge-triggered cell FDCE clocked by clk_out1_xlnx_clk_gen {rise@0.000ns fall@12.500ns period=25.000ns})
Destination: i_ariane/i_cva6/ex_stage_i/lsu_i/lsu_bypass_i/mem_q_reg[0][operation][3]/D
(rising edge-triggered cell FDCE clocked by clk_out1_xlnx_clk_gen {rise@0.000ns fall@12.500ns period=25.000ns})
```

The maximal clock period is $1/(\text{clock_period} - \text{slack})$, i.e. $1/(25.000 \cdot 10^{-9} - 0.094 \cdot 10^{-9}) = 40.15$ MHz in this illustration. Your actual critical path might be different from this one.

The synthesizer might cease optimizing the frequency when the slack is met. To overcome this, you can reduce the period until you get a “violated” slack³. The formula above still applies with a negative slack.

The actual board frequency is 20% lower than the maximal frequency of the reference design, so that you do not need to change the operating frequency⁴ even if the design maximal frequency decreases by 20%.

2.3 FPGA resources

In the unlikely case where tie persists when comparing frequencies (<1%), the results will be separated according to the FPGA resources (number of flip-flops + number of LUTs excluding LUTRAM) used by the whole FPGA (not only the CVA6 core).

2.4 CoreMark

CoreMark shall not be re-compiled.

For non-regression and basic RV32IM_Zicsr compatibility testing, it shall run successfully and return a result in simulation and on the board. For this year’s contest, we do not care about the CoreMark performance.

2.5 MNIST execution

MNIST shall not be re-compiled.

For non-regression and basic RV32IM_Zicsr compatibility testing, it shall run successfully and return the predicted digit 4, with the credence 82, both in simulation and on the FPGA board. For this year’s contest, we do not care about the MNIST performance.

3 Reporting the results

In both the 6-page report and in your recorded video, you’ll clearly report these results:

- Number of cycles and instructions of the FFT before and after your optimizations (and the corresponding change in %)
- Maximal frequency of your design before and after your optimizations (and the corresponding change in %)
- Resources used in your design (LUTs, flip-flops, BRAM, DSP...) of your top-level design, before and after your optimizations
- Correct operation of FFT, CoreMark and MNIST both in simulation and on the Zybo board.

³ Be careful to ensure the violation is on a critical path, typically on a setup timing. Other violations, like hold time violations, need to be fixed.

⁴ If you want to change the board frequency for other reasons, you need to modify `FPGA_UART_0_FREQUENCY` parameter in file `sw/bsp/config/fpga_platform_config.h`.

The report shall be written as a scientific paper, presenting your approach, your solution and commenting your results. It should help the jury to assess your solution.

The jury will double check in their environment the results of the teams who claim the best results in their reports. This also ensures that results are compared based on the same tool versions. So you'll publish a modified GitHub repository, as a fork of <https://github.com/thalesgroup/cva6-softcore-contest>. You'll create a **report** folder where you will upload:

- Your 6-page (max) report written as a scientific paper, in French or English, with illustrations⁵
- Simulation log (uart)
- P&R report with the maximum frequency (corev_apu/fpga/report_cva6_fpga_impl/cva6_fpga.timing.rpt)
- Report of resources (corev_apu/fpga/report_cva6_fpga_impl/cva6_fpga.utilization.rpt)
- Log of the execution on the FPGA board (copy or screenshot of the hyperterminal output) for FFT, CoreMark and MNIST.

We suggest that you keep your GitHub repository private until the end of the contest (May 4th, 2026, 23:59 CEST) and make it public right after. We'll look at the file timestamps to ensure you completed the contest on time.

Remember that your design shall run on the FPGA board and fulfill all constraints listed in **Annonce RISC-V contest 2025-2026 v1.3.pdf** (or its subsequent version).

Your presentation video shall last at most 10 minutes. All team members are encouraged to participate. The deadline to submit your video, a few days after May 5th, will be announced by the organizers. There is no special prize for the video, but it should be good enough to be presented to a large audience if your team is a winner. Make sure that you do not include others' copyrighted material (or anything else illegal in France).

During your recorded video, consider showing the execution of your solution on the Zybo board or executions on your machine. Because of the file size, do not upload the video in GitHub; you'll later get instructions for the file transfer.

A final word

If you find the contest difficult, remember that the other teams face the same level of difficulty. Even if you think your results are limited, submit them!

Version

These guidelines can undergo some evolutions based upon the teams' feedback. Please check their final version before submitting your results.

Date	Comment
2026-04-08	Initial version

⁵ An IEEE template is suggested: <https://www.ieee.org/conferences/publishing/templates.html>.